

Akash Iyer

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EDUCATION

Georgia Institute of Technology (GPA: 3.9/4.0)

M.S. Robotics

Atlanta, GA

Aug. 2025 – May 2027

University of Maryland (GPA: 3.8/4.0)

B.S. Mechanical Engineering

College Park, MD

Aug. 2022 – May 2025

EXPERIENCE

Mechanical Systems and Project Engineering Intern

May 2025 – Aug 2025, May 2026 – Aug 2026

Boeing

Ridley Park, PA

- Created CAD models of CH-47 transmission system components in CATIA to evaluate capability improvements
- Helped conduct testing of mechanical components to improve and solve issues with aircraft fuel system
- Prepared engineering change proposals to fulfill customer requirements and document engineering changes

Mechanical Engineering Undergraduate Teaching Fellow

Aug 2024 – May 2025

University of Maryland

College Park, MD

- Prepared material for and conducted midterm and final exam reviews for ENME 382: Materials Engineering
- Developed rubric and graded homework assignments in coordination with other Teaching Fellows
- Taught and assisted students with exam preparation and assignments at office hours

Undergraduate Researcher

Jan. 2024 – May 2025

University of Maryland

College Park, MD

- Developed and iterated on new and existing CAD models for a tuna-inspired AUV prototype
- Manufactured and integrated components using 3D printing, casting, and manual methods into the AUV prototype

Applications Engineering Intern

Jun. 2024 – Aug. 2024

Henkel

Bridgewater, NJ

- Analyzed and evaluated ideas from manufacturing to develop material formulations and manufacturing techniques
- Planned and executed tests of new formulations to ensure they can be made in the Henkel manufacturing network

Engineering Intern

Aug. 2023 – Jun. 2024

Gaia Energy Research Institute

Alexandria, VA

- Applied principles of physics, thermodynamics, and renewable energy systems to support design development
- Analyzed and developed computer models of physics, economics, and environmental impact of renewable energy

PROJECTS

Graph Search and Trajectory Optimization for Quadrupedal Footstep Planning

Jan. 2026 – Present

- Developed graph search and dynamic feasibility evaluation algorithms for quadrupedal footstep planning
- Modified simulation environments and evaluated various algorithms in Gazebo for the ANYmal and Unitree Go2
- Implemented algorithms on hardware, measuring graph construction time, failure modes, and goal completion rate
- Modified existing and developed new Github branches for multiple quadrupeds in QuadPips codebase

Turtlebot3 Autonomous Navigation Project

Aug. 2025 – Dec. 2025

- Implemented image processing and classification algorithm to identify target signs and turn accordingly
- Developed a state-machine to process LiDAR and odometry data to navigate to goal while avoiding obstacles
- Used ROS2 navigation stack and SLAM to map out a course, form a path, and traverse to waypoints in the course

Testudog Robotics Project | *Mechanical Subteam Lead*

Jan. 2023 – May 2025

- Developed structure and designed components from scratch for the chassis and legs of an autonomous robot dog
- Manufactured parts for robot through various methods such as waterjet, 3D printing, and metal cutting/bending
- Led mechanical team to coordinate design and manufacturing of robot and teach members engineering skills

TECHNICAL SKILLS

Electronics and Software: Raspberry Pi, Arduino, Python, C++, MATLAB, Pytorch, ROS 2, Gazebo, SLAM, Linux, Teamcenter IPDM, Github

CAD and Manufacturing: Solidworks, Fusion 360, Autodesk Inventor, CATIA, 3D Printing, Waterjet, CNC, Laser Cutting, Milling, Manual Machining